LABORATORY PROGRAM – 5

Implement Wordcount program on Hadoop framework

```
import java.io.Exception
import org.apache.hadoop.conf.Configured;
import org.apache.hadoop.fs.Path;
import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.FileInputFormat;
import org.apache.hadoop.mapred.FileOutputFormat;
import org.apache.hadoop.mapred.JobClient;
import org.apache.hadoop.mapred.JobConf;
import org.apache.hadoop.util.Tool;
import org.apache.hadoop.util.ToolRunner;

public class WCDriver extends Configured implements Tool {

    public int run(String[] args) throws IOException {

    if (args.length < 2) {
            System.out.println("Please give valid inputs");
            return -1;
    }

    JobConf conf = new JobConf(WCDriver.class);
    conf.setJobName("WordCount");

    FileInputFormat.setInputPaths(conf, new Path(args[0]));
    FileOutputFormat.setOutputPath(conf, new Path(args[1]));

    conf.setMapperClass(WCMapper.class);
    conf.setReducerClass(WCReducer.class);

    conf.setMapOutputKeyClass(Text.class);
      conf.setMapOutputValueClass(IntWritable.class);

    conf.setOutputKeyClass(Text.class);
```

```
        conf.setOutputValueClass(IntWritable.class);

        JobClient.runJob(conf);
        return 0;
        }

        // Main Method
        public static void main(String[] args) throws Exception {
        int exitCode = ToolRunner.run(new WCDriver(), args);
        System.out.println("Job Exit Code: " + exitCode);
        }
    }}
```

Mapper Code:

```
// Importing libraries
import java.io.IOException;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.LongWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.Mapper;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reporter;

public class WCMapper extends MapReduceBase
        implements Mapper<LongWritable, Text, Text, IntWritable> {

        // Map function
        public void map(LongWritable key, Text value,
                OutputCollector<Text, IntWritable> output,
                Reporter reporter) throws IOException {

        String line = value.toString();
```

```
        // Splitting the line on whitespace
        for (String word : line.split("\\s+")) {
        if (word.length() > 0) {
            output.collect(new Text(word), new IntWritable(1));
        }
        }
        }
}
```

Reducer Code

```
// Importing libraries
import java.io.IOException;
import java.util.Iterator;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;
import org.apache.hadoop.mapred.MapReduceBase;
import org.apache.hadoop.mapred.OutputCollector;
import org.apache.hadoop.mapred.Reducer;
import org.apache.hadoop.mapred.Reporter;

public class WCReducer extends MapReduceBase
      implements Reducer<Text, IntWritable, Text, IntWritable> {

        // Reduce function
        public void reduce(Text key, Iterator<IntWritable> values,
                OutputCollector<Text, IntWritable> output,
                Reporter reporter) throws IOException {
```

```
        int count = 0;

        // Counting the frequency of each word
        while (values.hasNext()) {
          count += values.next().get();
        }

      output.collect(key, new IntWritable(count));
        }
}
```

thilak@kali: ~/wordcount

Session Actions Edit View Help

```
┌──(thilak㉿kali)-[~]
└─$ mkdir wordcount
cd wordcount

┌──(thilak㉿kali)-[~/wordcount]
└─$ nano WCDriver.java

┌──(thilak㉿kali)-[~/wordcount]
└─$ nano WCMapper.java

┌──(thilak㉿kali)-[~/wordcount]
└─$ nano WCReducer.java

┌──(thilak㉿kali)-[~/wordcount]
└─$ javac -classpath `hadoop classpath` -d . WCDriver.java WCMapper.java WCReducer.java
hadoop: command not found
error: invalid flag: .
Usage: javac <options> <source files>
use --help for a list of possible options

┌──(thilak㉿kali)-[~/wordcount]
└─$ export HADOOP_HOME=/home/thilak/hadoop
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin

┌──(thilak㉿kali)-[~/wordcount]
└─$ javac -classpath $(hadoop classpath) -d . WCDriver.java WCMapper.java WCReducer.java

┌──(thilak㉿kali)-[~/wordcount]
└─$ jar -cvf wordcount.jar *
added manifest
adding: WCDriver.class(in = 1821) (out= 926)(deflated 49%)
adding: WCDriver.java(in = 1234) (out= 474)(deflated 61%)
adding: WCMapper.class(in = 1762) (out= 708)(deflated 59%)
adding: WCMapper.java(in = 599) (out= 303)(deflated 49%)
adding: WCReducer.class(in = 1529) (out= 604)(deflated 60%)
adding: WCReducer.java(in = 590) (out= 284)(deflated 51%)

┌──(thilak㉿kali)-[~/wordcount]
└─$ nano input.txt

┌──(thilak㉿kali)-[~/wordcount]
└─$ hdfs dfs -mkdir /wcinput
hdfs dfs -put input.txt /wcinput/

┌──(thilak㉿kali)-[~/wordcount]
└─$ hadoop jar wordcount.jar WCDriver /wcinput /wcoutput
2026-05-02 14:51:19,497 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2026-05-02 14:51:19,697 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2026-05-02 14:51:19,698 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2026-05-02 14:51:19,741 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2026-05-02 14:51:20,005 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with ToolRunner to remedy this.
2026-05-02 14:51:20,251 INFO mapred.FileInputFormat: Total input files to process : 1
2026-05-02 14:51:20,286 INFO mapreduce.JobSubmitter: number of splits:1
2026-05-02 14:51:20,536 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_local999704516_0001
2026-05-02 14:51:20,537 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-05-02 14:51:20,777 INFO mapreduce.Job: The url to track the job: http://localhost:8080/
2026-05-02 14:51:20,785 INFO mapreduce.Job: Running job: job_local999704516_0001
2026-05-02 14:51:20,791 INFO mapred.LocalJobRunner: OutputCommitter set in config null
2026-05-02 14:51:20,812 INFO mapred.LocalJobRunner: OutputCommitter is org.apache.hadoop.mapred.FileOutputCommitter
2026-05-02 14:51:20,826 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:51:20,826 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:51:20,931 INFO mapred.LocalJobRunner: Waiting for map tasks
2026-05-02 14:51:20,942 INFO mapred.LocalJobRunner: Starting task: attempt_local999704516_0001_m_000000_0
2026-05-02 14:51:20,991 INFO output.FileOutputCommitter: File Output Committer Algorithm version is 2
2026-05-02 14:51:20,993 INFO output.FileOutputCommitter: FileOutputCommitter skip cleanup _temporary folders under output directory:false, ignore cleanup failures: false
2026-05-02 14:51:21,054 INFO mapred.Task:  Using ResourceCalculatorProcessTree : [ ]
2026-05-02 14:51:21,081 INFO mapred.MapTask: Processing split: hdfs://localhost:9000/wcinput/input.txt:0+40
2026-05-02 14:51:21,122 INFO mapred.MapTask: numReduceTasks: 1
```

```
                HDFS: Number of bytes read=80
                HDFS: Number of bytes written=30
                HDFS: Number of read operations=15
                HDFS: Number of large read operations=0
                HDFS: Number of write operations=4
                HDFS: Number of bytes read erasure-coded=0
        Map-Reduce Framework
                Map input records=3
                Map output records=7
                Map output bytes=68
                Map output materialized bytes=88
                Input split bytes=91
                Combine input records=0
                Combine output records=0
                Reduce input groups=4
                Reduce shuffle bytes=88
                Reduce input records=7
                Reduce output records=4
                Spilled Records=14
                Shuffled Maps =1
                Failed Shuffles=0
                Merged Map outputs=1
                GC time elapsed (ms)=12
                Total committed heap usage (bytes)=432013312
        Shuffle Errors
                BAD_ID=0
                CONNECTION=0
                IO_ERROR=0
                WRONG_LENGTH=0
                WRONG_MAP=0
                WRONG_REDUCE=0
        File Input Format Counters
                Bytes Read=40
        File Output Format Counters
                Bytes Written=30

┌──(thilak㉿kali)-[~/wordcount]
└─$ hdfs dfs -cat /wcoutput/part-00000
big     1
data    2
hadoop  2
hello   2

┌──(thilak㉿kali)-[~/wordcount]
└─$
```